

Guardians of Credit: “A Comprehensive Analysis of Machine Learning Models for Credit Risk Detection”

1st Tejanshu Bhandari

School of Computer Engineering
KIIT Deemed to be University
Bhubaneswar, Odisha, India
btejanshu@gmail.com

2nd Prasun Payne

School of Computer Engineering
KIIT Deemed to be University
Bhubaneswar, Odisha, India
prspayne@gmail.com

3rd Mahendra Kumar Gourisaria

School of Computer Engineering
KIIT Deemed to be University
Bhubaneswar, Odisha, India
mkgourisaria2010@gmail.com

4th Sudhansu Shekhar Patra

School of Computer Engineering
KIIT Deemed to be University
Bhubaneswar, Odisha, India
sudhanshupatra@gmail.com

5th Vishwanath Petli

School of Computer Engineering
KIIT Deemed to be University
Bhubaneswar, Odisha, India
vishalpetli73@gmail.com

Abstract—Credit cards are widely used in today’s digital and cashless world, however they operate at the risk of borrower to repay their credit loan, often times causing huge losses to the financial sector. Identifying such risks and preventing them beforehand is crucial. This paper presents a comprehensive study for credit risk detection by making use of different machine learning algorithms to classify customers into 'good' and 'bad' credit categories. This paper helps identify such risks and hence prevent them beforehand. The ANN model outperforms others in terms of the accuracy and ROC curve with an accuracy of 85.88%. Random Forest model provides the most balanced confusion matrix followed by confusion matrix of XG-Boost. Synthetic data points were generated to balance the imbalanced dataset, and feature engineering techniques is used to eliminate less relevant features. The study aims to help financial institutions reduce credit loss risk and identify creditworthy customers.

Keywords—Credit card, financial sector, Artificial Neural Networks, XG-Boost, and K-Nearest Neighbor, Credit Risk

I. INTRODUCTION

In the rapidly evolving digital economy, credit cards serve as a pivotal financial instrument, providing consumers with a predetermined line of credit for executing transactions. Issued predominantly by banking institutions, these cards facilitate the procurement of goods, services, and cash advances on credit. Credit cards have instigated a significant transformation in the economy too by substantially mitigating fraudulent transactions. They have played a pivotal role in curbing the circulation of black money, thereby preventing the emergence of a shadow economy in the market. This has not only augmented the transparency of financial transactions but also fortified the overall integrity of the economic system. Their widespread adoption in today's cashless society underscores their significance. Nevertheless, the credit extended by financial institutions is not immune to occasional losses stemming from non-repayment by borrowers.

The relentless progression in technology, particularly in the fields of machine learning and artificial intelligence, offers a beacon of hope in combating this issue of credit risk. Machine Learning finds its use in various different fields such as Loan Approval [1], Stock Prediction [2], and Finance applications [3]. Deep Learning is the foundation of Machine Learning and hence has also found many use cases as Stock market prediction [4] and Loan Approval [5]. The primary

objective of this paper is to discern customers who can be entrusted with credit by financial institutions, as well as those who pose a risk. This will help borrowers prevent credit risks and hence avoid a potential financial loss. This has been achieved by making use of different machine learning approaches such as Artificial Neural Network (ANN), Random Forest, Logistic Regression, K-Nearest Neighbor (KNN), Support Vector Machine (SVM), and XG-Boost. The efficacy of every model was substantiated by various metric results. The following section is partitioned as follows (II) Related work which contains valuable research work in the same field (III) Proposed Methods contain information about the dataset, data pre-processing, and the different algorithms used (IV) Result and Discussion contain information about the hardware and software used along with evaluation matrices for every algorithm and (V) Conclusion and future work, which discuss the future scopes in this field.

II. RELATED WORK

The importance of credit risk assessment in the banking and financial sector has been amplified by the 2007–2008 global financial crisis and the regulatory framework of Basel norms [6]. This has constituted a notable focus of research within the domains of finance and machine learning. Despite this extensive research and measures taken to prevent credit risks, it remains a challenging issue due to several factors. One of the primary factor occurs from infrequent default events and the need for longer timeframes to assess credit risk. This rarity of data required for building accurate credit risk models often leads to reliance on assumptions and alternative data sources. Another challenge is the absence of comprehensive public datasets, which can hamper machine learning models' ability to precisely forecast credit risk.

S. Shi et al. [7] (2022) provided a notable comprehensive review of major research contributions over the past eight years. The paper explores the utilization of statistical, machine learning, and deep learning methodologies to tackle credit risk challenges. It also introduced an innovative classification approach for machine learning-based credit risk algorithms and evaluated their performance through ranking analysis. In a similar review by S. Bhatore et al. [8] (2020) a comprehensive analysis of machine learning techniques used for credit risk evaluation was provided. The authors reviewed 136 papers, focusing on the impact of hyperparameters on

machine learning techniques and the limitations of current studies. Their findings reveal a rising trend in the utilization of Ensemble and Hybrid models, particularly those integrating neural networks and Support Vector Machines (SVM), for applications such as credit scoring. N. Rtayli et al [9] (2020) in their work made use of several ML algorithms such as SVM, iForest, LOF, Decision Tree for credit risk detection. It was observed that SVM based on RFC provided superior performance and achieved a sensitivity score of 87%. X. Wang et al. [10] (2022) applied an oversampling technique to handle imbalanced credit default datasets. The paper further compared the credit risk detection performance of different machine learning algorithms used.

A. Motwani et al. [11] (2018) presents an in-depth analysis of ensemble methods for credit risk detection. They applied 15 ML algorithms with two stages of classification and detection. Most the ML techniques garnered an accuracy of far less than 80 percent, and finally a predictive model for Credit Risk Detection based on ensemble technique was proposed. This proposed model was assessed on basis of various performance metrics, and it resulted in 81% prediction accuracy and was found to have a better correlation coefficient. In a separate investigation conducted by D. Swain et al. [12] (2021) credit score was predicted, making use of varied machine learning algorithms such as linear regression, K-Nearest Neighbor algorithm (KNN), and logistic regression. It was found KNN algorithm provided a very healthy accuracy of 89%.

III. PROPOSED METHOD

This section consists of - A) Dataset Description B) Data Pre-processing C) Method Description

A. Dataset Description

The data used for this project is from Kaggle [13]. The dataset contains details of 1,000 customers with the target variable being 'Class'. The data contains 20 features that determine the eligibility of a customer for credit and it could be used to predict if the customer could be given credit. This is a great dataset for predicting if the customer could be given credit with some data cleaning and feature engineering for building. The dataset is also notably imbalanced and raw demanding pre-processing to remove the outliers.

B. Data Pre-Processing

This section is divided into three major sections – 1) Scaling Data 2) Encoding Data 3) Balancing Data 4) Dropping Features

1) *Scaling data:* The dataset is pre-processed to remove the outliers and to balance the data. The outliers were removed using the Standard Scalar method. This method removes the outliers by removing the data points that are beyond the range of 3 standard deviations from the mean. Standard scalar calculates the z score of each value in the column and replaces the value with Nan if the z score is greater than 3 or less than -3. The standard scalar is considered good when the dataset has a normal distribution. The removal of outliers makes the data ready to be processed further and perform feature engineering on it.

2) *Encoding Data:* The first step was to identify all the categorical features in the data. Then, the features were transformed into numerical values using either label encoding or one hot encoding, depending on the type of the feature.

This was done because machine learning models require numerical inputs. Some features were custom-encoded, which means they were mapped to specific numerical values. This was done for ordinal categorical features that did not have a clear order or hierarchy. Finally the Target column was selected to be encoded into binary in which good was encoded as 1 and bad was encoded as 0.

3) *Balancing Data:* The Balancing using Adaptive Synthetic Sampling (ADASYN) was done in order to make samples equally distributed. The principle of ADASYN is to generate synthetic data points for the minority class based on the intricacy of learning each task. The degree of imbalance is calculated using the formula (1).

$$d = \frac{m_s}{m_l} \quad (1)$$

Where m_s denotes the number of minority samples and m_l denotes the number of majority samples. The difficulty of learning is measured by the ratio of majority neighbors among the nearest neighbors of each minority example. The higher the ratio, the harder the example is to learn, and the more synthetic data is generated for it.

4) *Dropping Features:* The data contained features that were not important for the model. These features were dropped using the chi-square test, correlation matrix, and mutual info classification. Chi-square is a statistical method that compares the observed and expected frequencies of different outcomes or categories. It can be used to test whether there is a significant difference or relationship between categorical variables. Table 1. gives a peek into Chi score obtained for some features from the dataset which helped in choosing the Features to be dropped.

TABLE I. CHISCORE

FEATURE	CHI SCORE
checking_status	1.02e-24
duration	1.69e-05
credit_history	4.91e-11
credit_amount	0.416
savings_status	2.35e-06
employment	0.0009
installment_commitment	0.0500

Correlation matrix can help you understand the patterns and trends in your data, and also identify which variables are important for your analysis. After these two tests a mutual classification method of feature classification was also used for feature selection. Mutual information classification is a method of feature selection that uses the concept of mutual information to measure the relevance of each feature to the target variable.

C. Method Description

1) *Support Vector Machine (SVM):* SVM [14, 15] optimally chooses the hyperplane with the maximum margin, commonly referred to as the optimal hyperplane. SVMs represent a class of machine learning algorithms adept at executing tasks such as classification and regression. They achieve this by identifying the optimal hyperplane that effectively segregates data into distinct classes or outputs. SVMs have many advantages over other learning algorithms,

such as high accuracy, robustness, scalability, and theoretical guarantees. Some of the challenges and limitations of SVMs include choosing the appropriate kernel and parameters, dealing with imbalanced data, handling large-scale and high-dimensional data, and interpreting the results. The decision function for linear SVM can be expressed as shown in equation 2.

$$w^T x + b = 0 \quad (2)$$

2) *K-Nearest Neighbor (KNN)*: KNN algorithm [16, 17] is a straightforward yet efficacious machine-learning technique applied to both classification and regression tasks. It operates based on the premise that data points with analogous features tend to possess similar labels or outputs. Despite its simplicity, KNN faces challenges such as high computational cost, sensitivity to noise and outliers, and difficulty in choosing the optimal 'k' value and distance metric. The algorithm has found wide-ranging applications across various domains, including text categorization, image recognition, bioinformatics, and sentiment analysis, demonstrating its versatility and robustness.

3) *Decision Tree (DT)*: A decision tree classifier [18, 19], composed of nodes and branches, uses each node for a decision or test, and each branch for an outcome or choice. The tree starts at the root node and ends at the leaf nodes where final predictions are made. It can be constructed using methods like ID3, C4.5, CART, and CHAID. The algorithm is proficient in managing both numerical and categorical data, addressing missing values and noisy data, while also adeptly capturing non-linear relationships among features. However, it can be prone to overfitting, sensitive to small data changes, biased towards features with more values, and computationally expensive to train and optimize.

4) *Random Forest (RF)*: The Random Forest [20, 21] is a versatile machine learning algorithm capable of executing tasks like classification and regression through the amalgamation of multiple decision trees. Each DT is trained on a randomized subset of data and features, fostering diversity and diminishing correlations among the individual trees. The final output of the random forest is predicated on the average of the individual trees. However, random forests also have some drawbacks, such as being prone to overfitting, being computationally expensive, and being difficult to interpret.

5) *X Gradient-boost (XGBoost)*: XGBoost [22, 23] is a popular and robust machine learning algorithm that utilizes gradient boosting to combine multiple decision trees into a single model. XGBoost is an open-source library that offers a fast and scalable implementation of gradient boosting, with support for parallel and distributed computing. XGBoost can handle various types of data and problems, such as regression, classification, and ranking.

6) *Artificial Neural Network (ANN)*: Artificial Neural Networks [24, 25] are biological neural network-inspired computational tools. It consists of a large number of interconnected nodes, which act like artificial neurons. It can perform complex tasks like learning, pattern recognition, data classification, and prediction. This technique is used to solve

problems that are almost impossible to solve using conventional algorithms. The weighted sum of inputs(z) into a neuron is calculated using equation 3.

$$z = \sum_{i=1}^n (w_i \times x_i) + b \quad (3)$$

Where z is the weighted sum, w_i represent weights associated with each input, x_i denotes input values and represent the bias term.

IV. RESULTS AND DISCUSSION

This segment is subdivided into the following segments:

A) Hardware and Software, and B) Results.

A. Hardware and software

The research used a workstation that had an 11th Gen Intel(R) Core(TM) i7-1165G7 @ 2.80GHz CPU processor and 16 GB RAM and operated on Windows 11 OS. Python version 3.11.1, in Vs Code IDE, and Scikit-Learn modules and keras were the tools used.

B. Results

Following the construction and manual parameter tuning of all models, each model underwent evaluation based on different criteria such as accuracy, confusion matrix analysis, and the generation of ROC curves. The Artificial Neural Network (ANN) model outperformed all others, achieving an accuracy of 85.88%, also demonstrating a superior performance in terms of the ROC curve. Subsequently, the SVM model demonstrated an accuracy of 85%. Other models, including Decision Tree, Random Forest, XG Boost, and K-Nearest Neighbors (KNN), achieved accuracies of 72%, 82%, 84%, and 74% respectively. These results are tabulated in Table II, with their respective recall score, precision score and F1 score.

The confusion matrix for each model provided valuable insights into the problem solution. The KNN model was particularly effective in predicting true positives over false positives, as evidenced by a recall score of 97% for class 0 (bad). A balance between false positives and true negatives was observed in the confusion matrices of the Decision Tree and Random Forest models, making them robust models for predicting both classes. The KNN model also achieved a precision score of 96% for class 1 (good), indicating a high percentage of correctly classified positive cases from the total predicted positive cases. Figure 1, Figure 2, Figure 3, Figure 4, Figure 5 and Figure 6 demonstrate the confusion matrix of RF, ANN, XG-Boost, DT, SVM, KNN respectively. Figure 7, Figure 8, Figure 9 and Figure 10 show the ROC curve of the ANN, RF, XG-Boost, DT, SVM, KNN respectively. Figure 11 provides a visual comprehensive comparison of all the ROC curves, allowing for an effective evaluation.

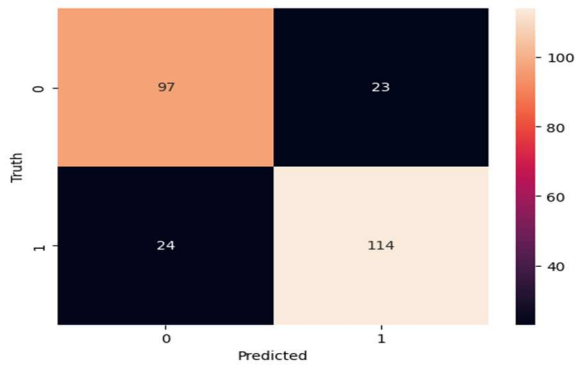


Fig. 1. Confusion Matrix of Random Forest

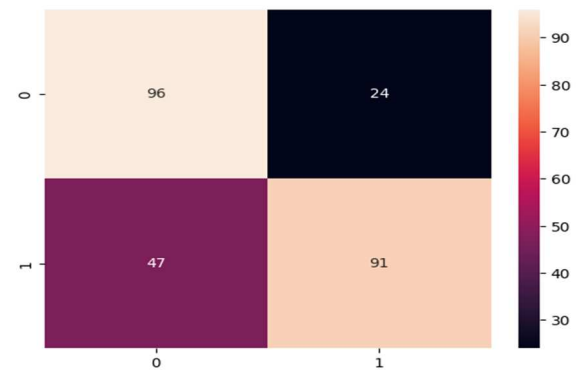


Fig. 4. Confusion Matrix of Decision Tree

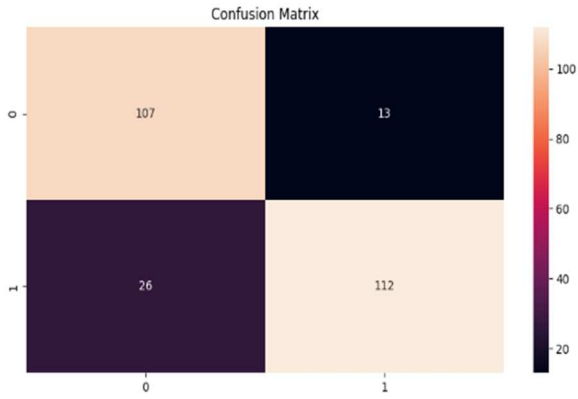


Fig. 2. Confusion Matrix of ANN

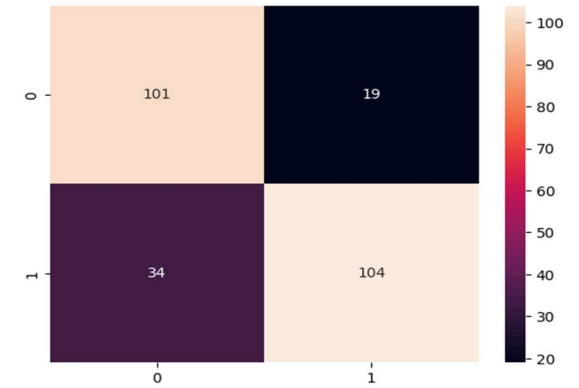


Fig. 5. Confusion Matrix of SVM

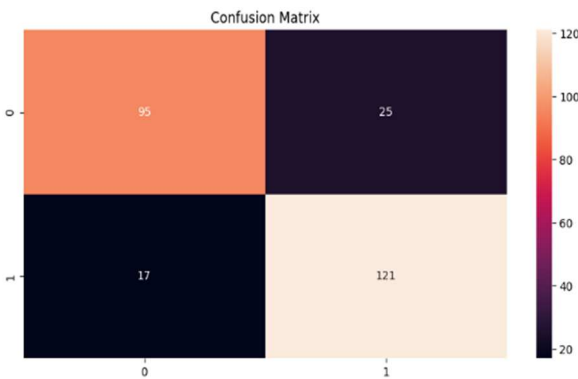


Fig. 3. Confusion Matrix of XG-Boost

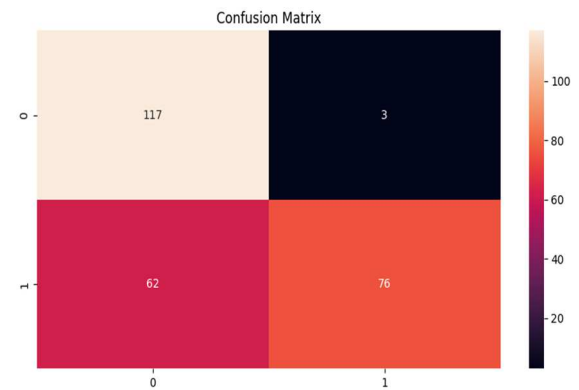


Fig. 6. Confusion Matrix of K-NN

TABLE II. METRICS TABLE

Model	Class	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	CM	
						TP	FP
						TN	FN
Decision Tree	0	72	67	80	73	96	24
	1		79	66	72	47	91
Random Forest	0	82	80	81	80	97	23
	1		83	83	83	24	114
SVM	0	85	80	89	85	101	19
	1		90	81	85	34	104
XG-Boost	0	84	85	85	82	95	25
	1		83	83	85	17	121
KNN	0	75	65	97	78	117	3
	1		96	55	70	62	76
ANN	0	85.88	80	89	85	107	13
	1		90	81	85	26	112

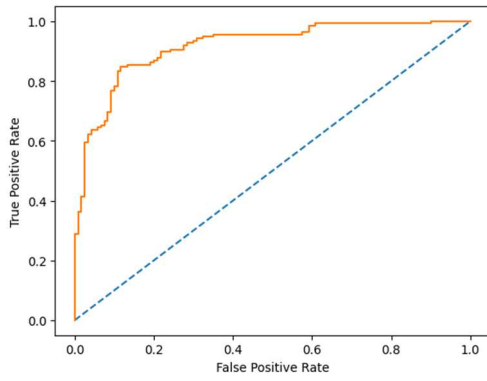


Fig. 7. ROC of ANN

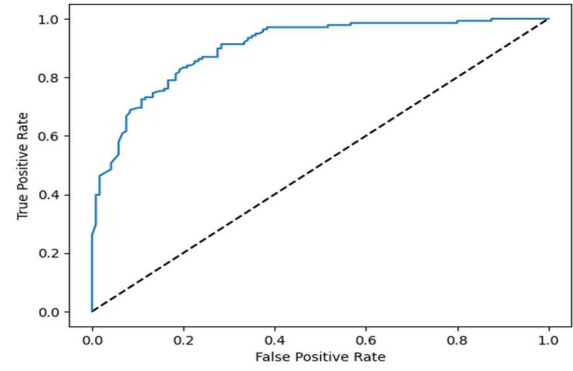


Fig. 9. ROC of Random Forest

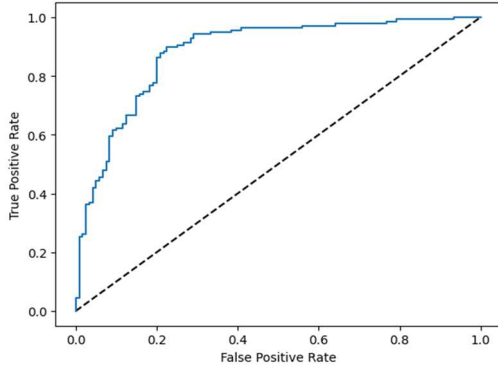


Fig. 8. ROC of XGBoost

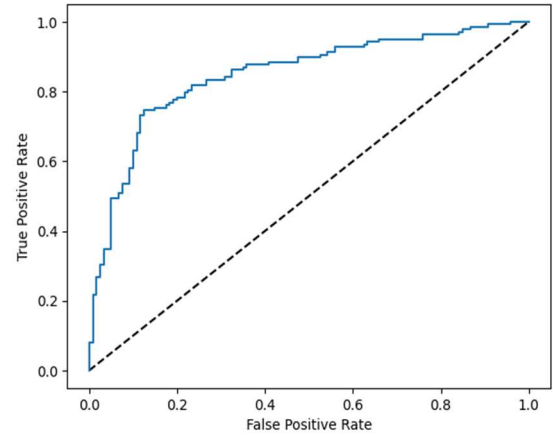


Fig. 10. ROC of SVM

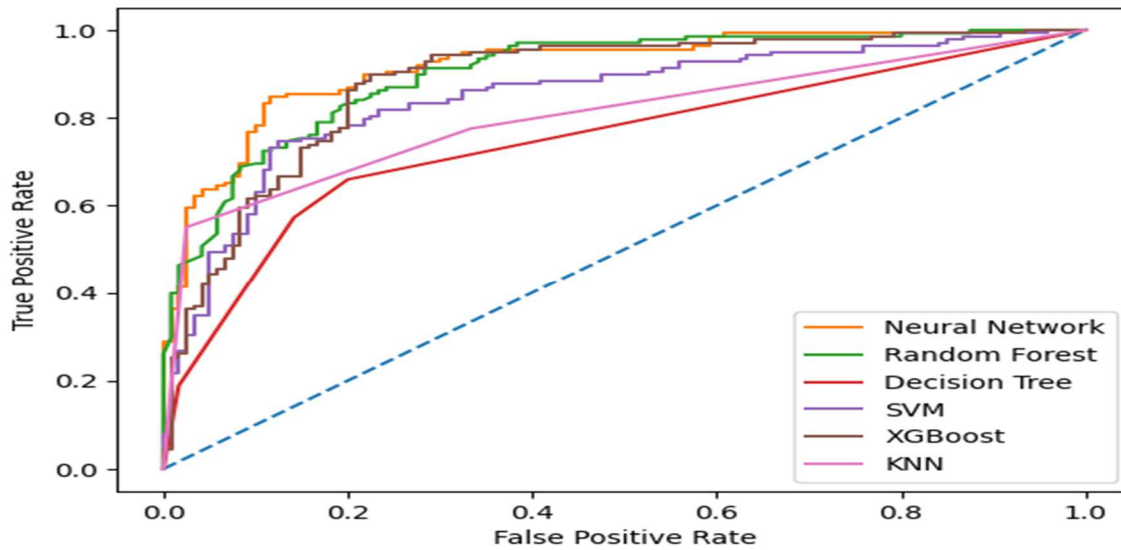


Fig. 11. Comparison of ROC curve of all algorithms

V. CONCLUSION AND FUTURE WORK

The Artificial Neural Network (ANN) model demonstrated the highest performance, achieving an accuracy of 85.88% and outperforming all other models in terms of the ROC curve. In close succession, the Support Vector Machine (SVM) model exhibited an accuracy rate of 85%. Other models, including Decision Tree, Random Forest, XG Boost, and K-Nearest Neighbors (KNN), also showed promising results with accuracies ranging from 72% to 84%. For future work, it would be advantageous to investigate alternative machine learning models and techniques to enhance the

accuracy of predictions even further. Additionally, incorporating more features or using feature selection techniques could potentially enhance the performance of the models. Lastly, it would be interesting to investigate how these models perform on different datasets or in different application domains.

ACKNOWLEDGMENT

The author would like to put forth gratitude to all the co-authors and especially to Professor M.K Gourisaria Sir for their insightful suggestions and supervision.

REFERENCES

- [1] A. P. Behera, S. S. Rautaray, M. Pandey, and M. K. Gourisaria, "Predictive loan approval model using logistic regression" in *Smart Computing Techniques and Applications: Proceedings of the Fourth International Conference on Smart Computing and Informatics*, Singapore: Springer, vol. 1, pp. 715-722, 2021.
- [2] N. Pahwa, N. Khalfay, V. Soni, and D. Vora, "Stock prediction using machine learning a review paper," *International Journal of Computer Applications*, vol. 163, no. 5, pp. 36-43, 2017.
- [3] M. F. Dixon, I. Halperin, and P. Bilokon, "Machine Learning in Finance," vol. 1170. Berlin/Heidelberg, Germany: Springer International Publishing, 2020.
- [4] S. K. Raipitam, S. Kumar, T. Dhanani, S. Bilgaiyan, and M. K. Gourisaria, "Comparative study on stock market prediction using generic CNN-LSTM and ensemble learning," in *International Conference on Network, Multimedia and Information Technology (NMITCON)*, IEEE, pp. 1-6, 2023.
- [5] Y. Abakarim, M. Lahby, and A. Attioui, "Towards an efficient real-time approach to loan credit approval using deep learning," in *9th International Symposium on Signal, Image, Video, and Communications (ISIVC)*, IEEE, pp. 306-313, 2018.
- [6] Basel Committee on Banking Supervision, *Credit Risk Modelling: Current Practices and Applications*. Basel: Bank for International Settlements. Retrieved from BIS website, 1999.
- [7] S. Shi, R. Tse, W. Luo, S. D'Addona, and G. Pau, "Machine learning-driven credit risk: A systemic review," *Neural Computing Applications*, vol. 34, no. 17, pp. 14327-14339, 2022.
- [8] S. Bhatore, L. Mohan, and Y. R. Reddy, "Machine learning techniques for credit risk evaluation: A systematic literature review," *Journal of Banking Financial Technology*, vol. 4, pp. 111-138, 2020.
- [9] N. Rtayli and N. Enneya, "Selection features and support vector machine for credit card risk identification," *Procedia Manufacturing*, vol. 46, pp. 941-948, 2020.
- [10] X. Wang, K. Zong, and C. Luo., "Credit risk detection based on machine learning algorithms," *International Journal of Financial Services Management*, vol. 11, no. 3, pp. 183-189, 2022.
- [11] A. Motwani, G. Bajaj and S. Mohane, "Predictive modelling for credit risk detection using ensemble method," *International Journal of Computer Sciences and Engineering*, vol. 6, no. 6, pp. 863-867, 2018.
- [12] D. Swain, R. Agrawal, A. Chandak, V. Lapshetwar, N. Chandak, and A. Vaswani, "Credit score prediction using machine learning" in *Machine Learning and Information Processing: Proceedings of ICMLIP 2020*, Singapore: Springer, pp. 69-81, 2021.
- [13] J. V. Rijn, (2023, February). credit_risk_customers. Retrieved September 19, 2023 from <https://www.kaggle.com/datasets/ppb00x/credit-risk-customers>
- [14] K. Dutta, S. Chandra, and M. K. Gourisaria, "Early-Stage detection of liver disease through machine learning algorithms" in *Advances in Data and Information Sciences*. Singapore: Springer, pp. 155-166, 2022.
- [15] S. Huang, N. Cai, P. P. Pacheco, S. Narrandes, Y. Yang and W. XU, "Applications of support vector machine (SVM) learning in cancer genomics," *Cancer Genomics Proteomics*, vol. 15, no. 1, pp. 41-51, 2018.
- [16] Y. Wang and Z. O. Wang, "A fast KNN algorithm for text categorization" in *international conference on machine learning and cybernetics*, vol. 6. IEEE, 2007.
- [17] U. Singh, M. K. Gourisaria and B. K. Mishra, "A dual Dataset approach for the diagnosis of hepatitis C virus using Machine Learning" in *IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, pp. 1-6, 2022.
- [18] G. M. Cramer, R. Ford, and R. L. Hall, "Estimation of toxic hazard—A decision tree approach," *Food Cosmetics Toxicology*, vol. 16, no. 3, pp. 255-276, 1978.
- [19] A. Sahu, G. M. Harshvardhan, and M. K. Gourisaria, "A dual approach for credit card fraud detection using neural network and data mining techniques" in *17th India council international conference (INDICON)*, IEEE pp. 1-7, 2020.
- [20] Y. Qi, "Random Forest for Bioinformatics." *Ensemble Machine Learning: Methods and Applications*, pp. 307-323, 2012.
- [21] M. K. Gourisaria, S. Khare, A. V. Patel, M. Sahni, and E. Leon-Castro, "A deep analysis of gilled mushroom edibility prediction using dimensionality reduction techniques" in *Novel Developments in Futuristic AI-Based Technologies*. Singapore: Springer Nature Singapore, pp. 67-92, 2023.
- [22] Y. Qiu, J. Zhou, M. Khandelwal, H. Yang, P. Yang, and C. Li, "Performance evaluation of hybrid WOA-XGBoost, GWO-XGBoost, and BO-XGBoost models to predict blast-induced ground vibration," *Engineering with Computers*, pp. 1-18, 2021.
- [23] I. Saha, M. K. Gourisaria, G. M. Harshvardhan, "Classification system for prediction of chronic kidney disease using data mining techniques" in *Advances in Data and Information Sciences: Proceedings of ICDIS*. Singapore: Springer Singapore, pp. 429-443, 2022.
- [24] S. Palani, S. Y. Liong, and P. Tklich, "An ANN application for water quality forecasting," *Marine Pollution Bulletin*, vol. 56, no. 9, pp. 1586-1597, 2008.
- [25] G. Jee, G. M. Harshvardhan and M. K. Gourisaria, "Juxtaposing inference capabilities of deep neural models over posteroanterior chest radiographs facilitating COVID-19 detection," *Journal of Interdisciplinary Mathematics*, vol. 24, no. 2, pp. 299-325, 2021.